

An Internship Report
Of
Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (CSE)



By

Roll Number of Student: **2410031596**

Name of Student: **Akshita Yadav**

Batch: **3CSE12**

CANDIDATE'S DECLARATION

I, **Akshita Yadav**, do hereby declare that the internship report titled “**Siemens Data Science Master Virtual Internship**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Engineering.

This report presents the knowledge, learning, activities, assessments, and outcomes obtained during the **8-Week AICTE–EduSkills Virtual Internship Program** in the domain of **Siemens Data Science Master**. The internship was undertaken from **June 2026 to August 2026**.

All the references and supporting materials used in the preparation of this report have been appropriately acknowledged. I affirm that this report represents my learning and understanding of the internship program and has not been submitted to any other institute for the fulfilment of any other degree or diploma requirement.

I shall be solely responsible for any kind of copyright violation or incorrect representation of information in this report.

Signature:

Student Name: Akshita Yadav

Roll No.: 2410031596

ACKNOWLEDGEMENT

I am using this opportunity to express my sincere gratitude to everyone who supported me throughout the internship program. I am thankful for their guidance, constructive feedback, encouragement, and support, which helped me complete the internship successfully and improve my understanding of Data Science and related technologies.

I would like to thank the **AICTE – EduSkills Virtual Internship Program team and EduSkills Academy** for providing me with the opportunity to participate in the **Siemens Data Science Master Virtual Internship**. The structured eight-week learning program provided me with an organized platform to develop industry-relevant knowledge and gain practical exposure in areas such as data engineering, machine learning, data preparation, applications and use cases, and platform administration.

I am also thankful to **IILM University, Greater Noida**, and the faculty members of the **School of Computer Science and Engineering** for their continuous support, encouragement, and guidance during the internship.

The week-wise structure of the internship, assessments, assignments, and final assessment helped me maintain a systematic approach towards learning. The program also allowed me to understand how concepts related to Data Science and Machine Learning can be connected with practical and industry-oriented applications.

Finally, I express my heartfelt gratitude to my parents and everyone who encouraged and supported me throughout this learning journey.

Signature:

Student Name: Akshita Yadav

Roll No.: 2410031596

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3. INTERNSHIP COMPLETION CERTIFICATE

The Internship Offer Letter issued by EduSkills confirms the selection of Akshita Yadav, B.Tech, Computer Engineering, IILM University, Greater Noida, for the 8-Week AICTE–EduSkills Virtual Internship Program in the Siemens Data Science Master domain for the period June 2026 to August 2026.

The internship documentation further specifies that successful completion required participation in the structured learning plan, weekly assessments, required project documentation and assignments, and the Final Assessment Test.

The relevant internship completion certificate and professional/master certifications are attached as supporting documents in the final report.

4. PROJECT DESCRIPTION

4.1 INTRODUCTION

This report presents a detailed account of the 8-week Siemens Data Science Master Virtual Internship undertaken under the AICTE–EduSkills Virtual Internship Program from June 2026 to August 2026 as part of the Bachelor of Technology program at IILM University, Greater Noida.

The primary objective of the internship was to develop a structured understanding of Data Science, Data Engineering, Machine Learning, Data Preparation, Applications and Use Cases, and Platform Administration through a week-wise learning program. The internship provided exposure to different stages of the Data Science lifecycle, beginning with applications and use cases and progressing through data engineering, Machine Learning, advanced data preparation, deployment, and platform administration.

The program combined theoretical learning with practical exposure and industry-relevant skills. Students completed weekly assessments, project documentation and assignments where applicable, followed by a Final Assessment Test. Overall, the internship helped strengthen technical knowledge, analytical abilities, and understanding of real-world Data Science workflows.

4.2 ORGANIZATION PROFILE

EduSkills Academy

EduSkills Academy focuses on developing industry-relevant skills among students through structured learning and internship programs. The AICTE–EduSkills Virtual Internship Program provides students with opportunities to gain exposure to various technology domains through virtual internship tracks, supplementing academic education with practical and industry-oriented learning. The selected domain for this internship was **Siemens Data Science Master**, which provided a structured learning path covering Data

Science and related technologies. The internship was organized into eight weeks, with each week dedicated to a specific professional or master-level module. It also included weekly assessments and a final credential validation/internship grade point assessment, providing an organized environment for students to develop and demonstrate their skills.

4.3 INTERNSHIP PROBLEM STATEMENT

In today's digital environment, organizations generate large volumes of data from various sources. The major challenge is transforming this data into meaningful information that can support decision-making and intelligent applications.

Students often study databases, data processing, Machine Learning, and Artificial Intelligence as separate subjects. However, understanding how these concepts work together in an industry-oriented Data Science workflow requires structured exposure.

The internship addressed this by developing a systematic understanding of the Data Science workflow, including identifying AI and Machine Learning problems, accessing and preparing data, learning foundational and advanced Machine Learning techniques, understanding data handling and automation, and gaining exposure to model deployment, monitoring, and platform administration. The internship covered Applications & Use Cases, Data Engineering, Machine Learning, Data Preparation, and Platform Administration through professional and master-level modules.

4.4 INTERNSHIP OBJECTIVES

The major objectives of the internship were:

- To understand the fundamental concepts and applications of Data Science.
- To identify business problems that can be addressed using Artificial Intelligence and Machine Learning.
- To understand how data can be accessed, loaded, transformed, and calculated.
- To develop foundational knowledge of Machine Learning.

- To understand advanced data preparation and handling techniques.
- To understand advanced Machine Learning techniques.
- To understand the application and deployment of Machine Learning models.
- To understand methods for monitoring Machine Learning models.
- To gain knowledge of platform configuration and administration.
- To understand the role of enterprise platforms in supporting Data Science and Artificial Intelligence workflows.
- To develop industry-relevant technical skills.
- To improve analytical and problem-solving abilities.
- To complete weekly assessments and required assignments successfully.
- To successfully complete the Final Credential Validation / Internship Grade Point Assessment.
- To obtain professional and master-level certifications associated with the internship.

The internship offer letter states that the program was designed to provide structured learning combined with practical exposure and that the successful completion of required assessments leads to the award of the Virtual Internship Certificate.

4.5 SCOPE OF THE INTERNSHIP

The scope of this internship covers major areas of the Data Science ecosystem. It includes Applications and Use Cases, focusing on the use of Artificial Intelligence and Machine Learning for solving real-world problems. Data Engineering covers accessing, loading, transforming, and working with data. Machine Learning includes foundational and advanced concepts for learning patterns from data. Data Preparation focuses on advanced data handling and automation to support effective analysis. Applications & Use Cases Master introduces Machine Learning model deployment and monitoring. Finally, Platform Administration Master covers platform configuration and

management for Data Science and AI workflows. The internship was conducted virtually and emphasized structured learning, assessments, assignments, and practical exposure.

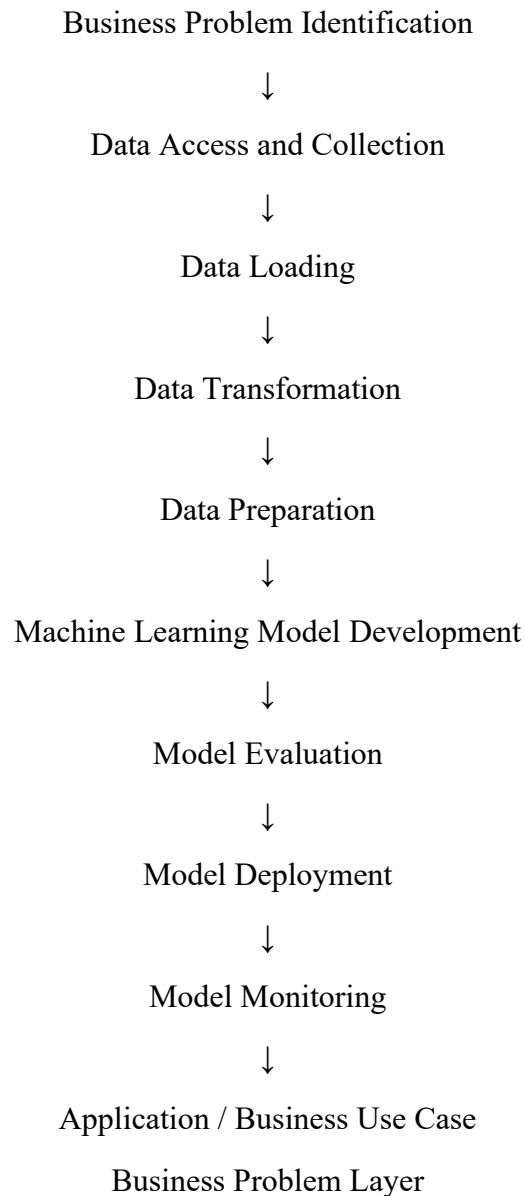
4.6 TECHNOLOGIES AND TOOLS USED

The internship covered major technologies, concepts, and platforms related to Data Science and Artificial Intelligence.

1. Data Science: Data Science combines data processing, statistical analysis, Machine Learning, and domain knowledge to obtain useful insights from data.
2. Data Engineering: Data Engineering focuses on accessing, loading, transforming, and preparing data for analysis and Machine Learning.
3. Machine Learning: Machine Learning involves developing models that learn patterns from data for tasks such as prediction and classification.
4. Data Preparation: Data Preparation involves organizing and transforming raw data into a suitable format for analysis and Machine Learning.
5. Artificial Intelligence Applications: AI and Machine Learning can be applied to different business and real-world problems depending on the data and desired outcomes.
6. Model Deployment and Monitoring: This covers concepts related to deploying and monitoring Machine Learning models.
7. Platform Administration: This includes platform configuration and management, particularly RapidMiner AI Hub.
8. Assessments and Learning Platform: Weekly assessments, project documentation, assignments where applicable, and a final assessment were used to evaluate learning progress.

4.7 DATA SCIENCE WORKFLOW / SYSTEM ARCHITECTURE

The overall learning architecture of the internship can be represented through the following stages:



System Architecture

1. Problem Identification Layer: The process begins by identifying a business or real-world problem suitable for AI or Machine Learning.

2. Data Engineering Layer: Relevant data is accessed, loaded, managed, and transformed using Data Engineering techniques.
3. Data Preparation Layer: Raw data is transformed and prepared for analysis or Machine Learning.
4. Machine Learning Layer: Prepared data is used to build Machine Learning models through professional and master-level learning.
5. Application Layer: Developed models are applied to relevant business or real-world use cases.
6. Deployment and Monitoring Layer: Models are deployed and monitored as introduced in the Applications & Use Cases Master module.
7. Platform Administration Layer: AI and Data Science workflows require platform configuration and administration, covered during the Platform Administration Master modules.

4.8 METHODOLOGY

The internship followed a structured week-wise methodology. Each week focused on a particular module, allowing the learning process to progress from fundamental concepts to more advanced topics.

The internship structure consisted of eight learning weeks followed by final credential validation and internship grade point assessment.

The methodology included:

1. Week-wise conceptual learning.
2. Practical exposure to relevant concepts.
3. Completion of weekly assessments.
4. Completion of required assignments and project documentation.
5. Progression from professional-level to master-level modules.
6. Final credential validation.
7. Internship Grade Point Assessment.

The official internship offer letter states that the final grade was based on overall performance, including weekly assessments, project submissions, and the final test.

4.9 WEEK-WISE INTERNSHIP ACTIVITIES

Week	Module	Key Learning / Activities
Week 1	Applications & Use Cases Professional	Identified business problems suitable for AI and Machine Learning and understood their real-world applications.
Week 2	Data Engineering Professional	Learned data accessing, loading, transforming, calculating, and preparing data for analysis and Machine Learning.
Week 3	Machine Learning Professional	Developed foundational Machine Learning skills and understood models, workflows, and real-world applications.
Week 4	Data Preparation Master	Learned advanced data handling, transformation, preparation, and automation techniques.
Week 5	Machine Learning Master	Studied advanced Machine Learning techniques, modeling concepts, and the relationship between data preparation and model performance.
Week 6	Applications & Use Cases Master	Learned practical Machine Learning applications, including model deployment and monitoring.
Week 7	Platform Administration Master	Learned platform configuration and management with exposure to RapidMiner AI Hub and enterprise AI environments.
Week 8	Platform Administration Master	Studied enterprise-level platform operations, management, administration, and reliable AI infrastructure.
Final Assessment	Credential Validation / Internship Grade Point Assessment	Completed weekly assessments, applicable project submissions/assignments, and the final assessment for internship completion and certification.

4.10 EXPECTED OUTCOMES AND LEARNING OUTCOMES

The internship provided valuable technical, analytical, and professional learning outcomes. I developed an understanding of Data Science, Data Engineering, Artificial Intelligence, and Machine Learning, including data access, transformation, preparation, advanced Machine Learning techniques, model deployment, and monitoring. I also

gained exposure to AI platform configuration, administration, and enterprise-level operations. The internship improved my ability to approach problems through a structured workflow: **Problem → Data → Preparation → Model → Application → Deployment → Monitoring**. In addition, it strengthened my self-learning ability, time management, problem-solving skills, technical documentation, assessment preparation, and understanding of industry-oriented learning. Overall, the internship provided a systematic foundation for working with data-driven technologies.

4.11 CERTIFICATIONS AND COMPLETION PROOF

During the internship, multiple professional and master-level certifications were obtained.

The uploaded certificates verify the following credentials:

Professional Certifications

1. Applications & Use Cases Professional Certification
2. Data Engineering Professional Certification
3. Machine Learning Professional Certification

Master Certifications

4. Data Engineering Master Certification
5. Machine Learning Master Certification
6. Applications & Use Cases Master Certification
7. Platform Administration Master Certification

These certificates demonstrate the completion of learning modules in different areas of Data Science and related technologies.

The internship offer letter also confirms that the program consisted of eight weeks of structured learning followed by final credential validation and internship grade point assessment.

The original certificates should be attached in the Annexure section of the final printed report.

5. SKILLS DEVELOPED DURING THE INTERNSHIP

- **Data Science:** Understood the overall Data Science workflow.
- **Data Engineering:** Learned data access, loading, transformation, and processing.
- **Machine Learning:** Developed foundational and advanced Machine Learning knowledge.
- **Data Preparation:** Learned data preparation and transformation techniques.
- **AI Applications:** Understood AI applications in real-world problems.
- **Deployment & Monitoring:** Gained exposure to model deployment and monitoring.
- **Platform Administration:** Learned AI platform management and enterprise operations.

6. CHALLENGES AND LEARNING EXPERIENCE

During the internship, a major challenge was understanding the relationship between different areas of Data Science, including data engineering, preparation, Machine Learning, deployment, monitoring, and platform management. The progressive week-wise structure helped connect these concepts effectively. The transition from professional to master-level modules further strengthened understanding. Weekly assessments encouraged consistent learning and improved overall technical knowledge.

7. CONCLUSION

The Siemens Data Science Master Virtual Internship provided a structured understanding of Data Science and related technologies. The eight-week program covered Applications & Use Cases, Data Engineering, Machine Learning, Data Preparation, advanced Machine Learning, model deployment and monitoring, and Platform Administration. It helped develop technical knowledge, analytical thinking, problem-solving, and self-learning abilities. The successful completion of professional and master-level modules resulted in certifications across Applications & Use Cases, Data Engineering, Machine Learning, and Platform Administration. Overall, the internship strengthened my understanding of the broader Data Science, Artificial Intelligence, and Machine Learning ecosystem.

8. FUTURE SCOPE

The knowledge gained during the internship provides a foundation for further development in Data Science and Artificial Intelligence. Future learning can include advanced Machine Learning, Deep Learning, real-world datasets, data engineering, cloud platforms, model deployment, monitoring, MLOps, Big Data, and AI-powered applications. The internship therefore provides a strong foundation for further specialization in Data Science, Machine Learning, Artificial Intelligence, Data Engineering, and MLOps.

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